

WASP-12b

R-0396

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230119 · created 2026-07-19T02:52:38+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459962.7984 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.02
Transit depth (Rp/Rs)^2	1.28 +/- 0.45 [%]
Orbital Inclination (inc)	82.65 +/- 2.55
Airmass coefficient 1 (a1)	1.028 +/- 0.011
Airmass coefficient 2 (a2)	-0.024 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#1 - [407, 229]
Optimal Aperture	4.56
Optimal Annulus	4.95
Transit Duration (day)	0.1219 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.113 ± 0.02 vs 0.1178 (deviation 0.24σ)
Duration fit vs published	0.1219 d vs 0.125 d (deviation 0.5σ)
Mid-time O–C	-1573.8 min
Depth SNR	5.7
Residual scatter	1.07 %

Light curve

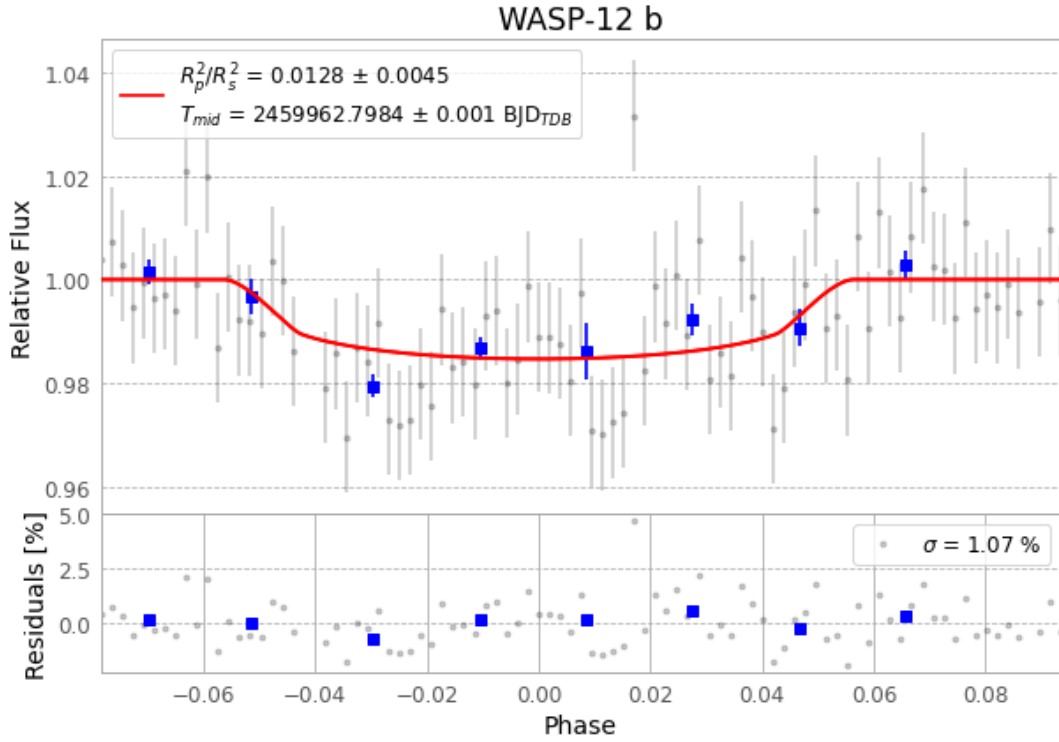


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-17.png (SHA-256 9c5b9310900589d1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [425, 198], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 833f89dfc45885e8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

95 FITS frames (62 MB), WASP-12230118045715.FITS → WASP-12230118093915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230119.log (47d71e577b78...), FinalLightCurve_WASP-12 b_2023-01-17.png (9c5b93109005...), FinalLightCurve_WASP-12 b_2023-01-17.csv (7381b2b46ef5...)

Compute resources

Wall time 390.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 02:52Z.